

# 4.1 Concept of Data Analysis

ڈیٹا کا تجزیہ یعنی اعداد کو ترتیب دے کر ان سے فیصلے کے قابل معلومات نکالنا۔

CS XII

Chapter 4: Data and Analysis

Lecture 01 of 8

## 4.1 Concept of Data Analysis

### LEARN IT

**Data analysis** is examining, organising and interpreting data so you can extract useful information. Schools study results, shops study sales, hospitals study patient records — all to make a decision, spot a trend or solve a problem.

Python is popular for this work because it is readable, talks to files and databases, does statistics, and draws graphs. Five lines of Pandas often replace an hour of spreadsheet clicking.

### WORKED EXAMPLE

A Hyderabad college has 400 CS marks in a spreadsheet. Analysis answers: What is the average? How many failed? Is the second shift weaker than the first? The numbers are data; those answers are information.

### اردو میں سمجھیں

ڈیٹا کا تجزیہ یعنی اعداد کو ترتیب دے کر ان سے فیصلے کے قابل معلومات نکالنا۔ پائٹھن اس لیے منتخب کیا جاتا ہے کہ وہ فائلیں، ڈیٹا بیس، حساب اور گراف سب سنبھال لیتا ہے۔

### CHECK YOURSELF

1. Give one school use of data analysis.
2. Name two Python superpowers for analysis.

Answers: 1. Study results to improve teaching (or similar). 2. Any two: read CSV/Excel/databases, organise tables, statistics, charts.

## Extra board practice

These MCQs and questions are extra to the textbook. Close the notes, attempt them, then open the key.

**MCQ 1.** Data analysis is:

- a. Examining data to extract useful information
- b. Only collecting rumours
- c. Only drawing logos
- d. Only cabling

**MCQ 2.** Python is used because:

- a. Files, databases, stats, plots in one language
- b. It is analog
- c. It cannot loop
- d. It cannot import

**MCQ 3.** Raw marks vs "class average 62":

- a. Data vs information
- b. Information vs data
- c. Both keys
- d. Both NAND

**MCQ 4.** A decision is the:

- a. Point of analysis
- b. First NAND
- c. Last OSI layer always
- d. Sprint name

**MCQ 5.** Without a question, analysis is:

- a. Aimless charting
- b. Always AI
- c. Always IoT
- d. Always HCI

**MCQ 6.** XII tools for analysis include:

- a. sqlite3, pandas, matplotlib, statistics
- b. Only LogiSim
- c. Only Figma
- d. Only Access macros

## Extra short questions

1. Define data analysis with a college example.
2. Four Python superpowers listed in the notes.
3. What is not analysis?
4. Link to XI digital literacy.

## Extra long question

1. A Hyderabad college has 400 CS marks. Write four analysis questions and which Python tool answers each.

## Extra answer key

**MCQs: 1. A 2. A 3. A 4. A 5. A 6. A**

**Short**

1. Process marks to decide who needs extra class.
2. Read files, talk to DB, statistics, charts (and readable syntax).
3. Dumping 400 rows on a slide with no summary.
4. Same collect–present loop, now with code.

**Long**

1. Average: pandas mean. Fail count: filter. Shift compare: groupby. Trend: matplotlib line if by year. Mention cleaning NaNs first.

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Next lecture: 4.1.1 – 4.1.2 Data Sources and Database Connection